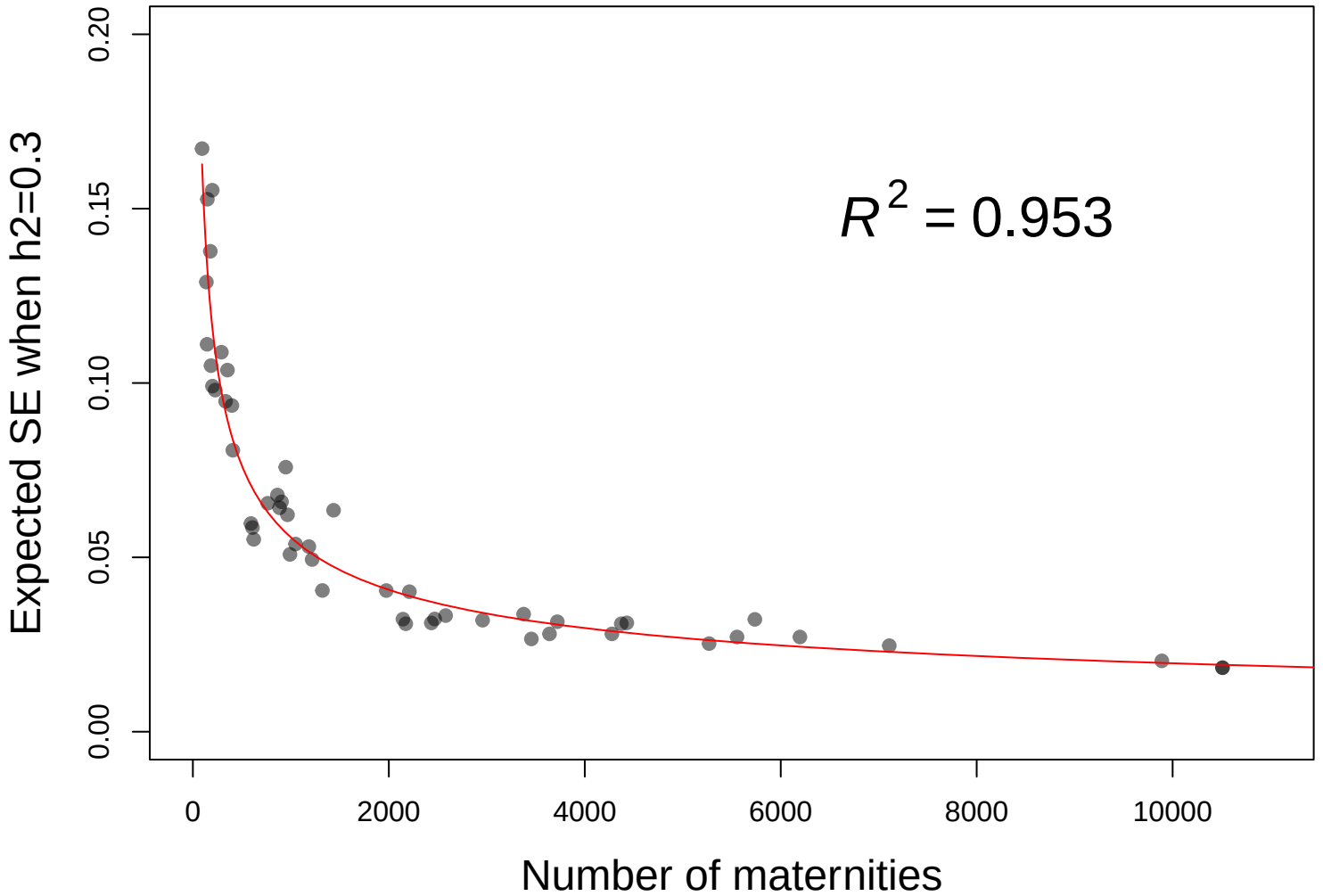
a)

Metric	R^2
records	0.938
maternities	0.953
paternities	0.901
full sibs	0.61
maternal sibs	0.694
maternal half sibs	0.478
paternal sibs	0.623
paternal half sibs	0.427
maternal grandmothers	0.422
maternal grandfathers	0.437
paternal grandmothers	0.45
paternal grandfathers	0.412
maximum pedigree depth	0.089
founders	0.524
mean maternal sibship size	0.242
mean paternal sibship size	0.171
non−zero F	0.062
F > 0.125	0.003
mean pairwise relatedness	0.19
pairwise relatedness>=0.125	0.21
pairwise relatedness>=0.25	0.323
pairwise relatedness>=0.5	0.463
var pairwise relatedness	0.329

b)